

Advanced Econometrics Lecture 04

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Today's agenda

1. Quick intro

- All you have to know to pass this course
- Ordered logit and ordered probit
- Marginal effects
- Proportional odds assumption
- Diagnostic tests for ordered choice models

2. Details

- Nice to know
- The stuff we usually teach our students majoring in econometrics
- Derivation of marginal effects
- Last homework solution
- Advanced Econometrics (course in Polish, different lecturer) exam exercise

Ordered Choice Models

- Dependent variable is a discrete and has more than two levels.
- y_i takes values in the set $\{0, 1, 2, \dots, J\}$, where J is a known constant.
- y_i codes states, and they can be ordered in a natural way.
- Those states do not have quantitative interpretation. If we consider final mark from a course, then mark "2" is not two times worse than mark "4".
- Observed variable y_t values are determined by latent variable y_i^*

$$y_i^* = x_i\beta + \epsilon_i$$

Ordered Choice Models

- Relationship between y_i and its hidden counterpart y_i^*

$$\left\{ \begin{array}{ll} y_i = 0 & \text{if } y_i^* \leq \alpha_1 \\ y_i = 1 & \text{if } \alpha_1 < y_i^* \leq \alpha_2 \\ y_i = 2 & \text{if } \alpha_2 < y_i^* \leq \alpha_3 \\ y_i = 3 & \text{if } \alpha_3 < y_i^* \leq \alpha_4 \\ \vdots & \vdots \\ y_i = J & \text{if } y_i^* > \alpha_J \end{array} \right.$$

- where J -**unknown** parameters $\alpha_1 < \alpha_2 < \dots < \alpha_J$ are called cutoff points.

Ordered Choice Models

$i = 0$

- $y_i = 0$ if $y_i^* \leq \alpha_1$
- $P(y = 0|x) = P(y^* \leq \alpha_1|x) = F(\alpha_1 - x\beta)$

$i = 1$

- $y_i = 1$ if $\alpha_1 < y_i^* \leq \alpha_2$
- $P(y = 1|x) = P(\alpha_1 \leq y^* \leq \alpha_2|x) = F(\alpha_2 - x\beta) - F(\alpha_1 - x\beta)$

$i = J$

- $y_i = J$ if $y_i^* > \alpha_J$
- $P(y = J|x) = P(y^* > \alpha_J|x) = 1 - F(\alpha_J - x\beta)$

Why?

Exercise 2¹

Let's analyse variable coding health status of respondents (*hstatus*). The variable takes three values: 1 for a person with bad health, 2 for moderate health, and 3 for good health status. Covariates for the dependent variable are: *income*, *female* – a binary variable (1 for women) and *num* – family size.

¹This exercise uses *randdata.dta* dataset from A. Colin Cameron and Pravin K. Trivedi, *Microeconometrics: Methods and Application*, Cambridge University Press, New York, May 2005.

polr from the MASS package

Re-fitting to get Hessian

Call:

```
polr(formula = health ~ income + female + num, data = rd)
```

Coefficients:

	Value	Std. Error	t value
income	0.0001159	0.00001112	10.424
female	-0.1410198	0.00280972	-50.190
num	0.0294082	0.01193174	2.465

Intercepts:

	Value	Std. Error	t value
1 2	-2.5789	0.0032	-802.7891
2 3	-0.5273	0.0469	-11.2546

Residual Deviance: 10588.14

AIC: 10598.14

ologit.reg function

```
> ologit = ologit.reg(health~income+female+num, data=rd)
Ordered Logit Regression
Log-Likelihood: -5294.072
No. Iterations: 5
McFadden's R2: 0.02912101
AIC: 10598.14
```

	Estimate	Std. error	t value	Pr(> t)	
income	0.0001157821	0.0000070506	16.4217	< 2.2e-16	***
female	-0.1412207117	0.0535073206	-2.6393	0.008308	**
num	0.0293452983	0.0126663105	2.3168	0.020515	*

----- Threshold Parameters -----

	Estimate	Std. error	t value	Pr(> t)	
Threshold (1->2)	-2.580795	0.092750	-27.8254	< 2.2e-16	***
Threshold (2->3)	-0.528933	0.077969	-6.7839	1.17e-11	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Marginal effects from margins.oglmx()

Marginal Effects on Pr(Outcome==1)

	Marg. Eff	Std. error	t value	Pr(> t)	
female	0.00404525081	0.00153378879	2.6374	0.008354	**
income	-0.00000334106	0.00000025207	-13.2545	< 2.2e-16	***
num	-0.00084680153	0.00036781053	-2.3023	0.021320	*

Marginal Effects on Pr(Outcome==2)

	Marg. Eff	Std. error	t value	Pr(> t)	
female	0.01780157241	0.00671462027	2.6512	0.008021	**
income	-0.00001466399	0.00000088939	-16.4877	< 2.2e-16	***
num	-0.00371662855	0.00160455352	-2.3163	0.020542	*

Marginal Effects on Pr(Outcome==3)

	Marg. Eff	Std. error	t value	Pr(> t)	
female	-0.0218468232	0.0082306515	-2.6543	0.007947	**
income	0.0000180050	0.00000010556	17.0560	< 2.2e-16	***
num	0.0045634301	0.0019691219	2.3175	0.020477	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Marginal effects from `ome()`

```
> x = c(7000, 0, 4)
```

```
> ome(model, x)
```

	alternative1	alternative2	alternative3	at X=
income	-0.000003274	-0.000021045	0.000017771	7000
female!	0.004257635	0.018310489	-0.022568124	0
num	-0.000830393	-0.005338129	0.004507737	4

(!) indicates marginal effect was calculated for discrete change of dummy variable from 0 to 1

Proportional odds model

Example 1 cont.

Let's assume that we observe variable y_i that codes customers' satisfaction from a certain product.

$$y_i \in \{\text{dissatisfied, neutral, satisfied}\} \quad (1)$$

Let's assign values (0, 1, 2) to the customers' satisfaction levels.

$$y_i = \beta \text{age}_i + \varepsilon_i \quad \varepsilon_i \sim N(0, 1)$$

$$\Pr(y_i \leq j) = \frac{\exp(\alpha_j - \beta \text{age})}{1 + \exp(\alpha_j - \beta \text{age})}$$

$$\text{Odds}(y_i | \text{age}) = \frac{\exp(\alpha_j - \beta \text{age})}{1 + \exp(\alpha_j - \beta \text{age})} / \frac{1}{1 + \exp(\alpha_j - \beta \text{age})} = \exp(\alpha_j - \beta \text{age})$$

$$\frac{\text{Odds}(y_i | \text{age} + 1)}{\text{Odds}(y_i | \text{age})} = \frac{\exp(\alpha_j - (\beta + 1)\text{age})}{\exp(\alpha_j - \beta \text{age})} = e^{-\beta} \quad (2)$$

Proportional odds assumption

$$Pr(y \leq m|x) = F(\alpha_m - x\beta) \quad \text{for } m = 1 \text{ to } J - 1 \quad (3)$$

This equation shows that the ordinary response model is equivalent to $J - 1$ binary regressions with the critical assumption that the slope coefficients are identical across each regression.

The parallel regression assumption implies that

$$\beta_1 = \beta_2 = \dots = \beta_{J-1}. \quad (4)$$

To the degree that the parallel regression assumption holds, the coefficients $\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_{J-1}$ should be close.

How to test the assumption?

J.S. Long, J. Freese, *Regression Models for Categorical Dependent Variables Using Stata*, Stata Press.

Proportional odds assumption

```
rat = read.csv(file="Ratings.csv", header=TRUE, sep=",")
View(cbind(rat$rating83c, rat$ia83, rat$dia))

rr = rep(0, 98)
rr[which(rat$rating83c=="BA_B_C")]=2
rr[rat$rating83c=="BAA"]=3
rr[rat$rating83c=="AA_A"]=4
rr[rat$rating83c=="AAA"]=5

# Estimate ordered logit for rating83c.
# polr from MASS package
ologit = polr(as.factor(rr)~ia83+dia, data=rat)
summary(ologit)
```

Proportional odds assumption - Brant's test

```
> brant(ologit)
```

```
-----  
Test for      X2      df    probability  
-----
```

```
Omnibus      68.74      4      0  
ia83         14.24      2      0  
dia          1.1       2     0.58  
-----
```

H0: Parallel Regression Assumption holds

```
      X2      df    probability  
Omnibus 68.742331  4    4.182557e-14  
ia83    14.237080  2    8.099485e-04  
dia     1.099094  2    5.772113e-01
```

Ordered probit assumptions - Greene p. 831

The model is built around a latent regression in the same manner as the binomial probit model. We begin with

$$y^* = \mathbf{x}'\beta + \varepsilon. \quad (5)$$

As usual, y^* is unobserved. What we observe is

$$\left\{ \begin{array}{ll} y_i = 0 & \text{if } y_i^* \leq \alpha_1 \\ y_i = 1 & \text{if } \alpha_1 < y_i^* \leq \alpha_2 \\ y_i = 2 & \text{if } \alpha_2 < y_i^* \leq \alpha_3 \\ y_i = 3 & \text{if } \alpha_3 < y_i^* \leq \alpha_4 \\ \vdots & \vdots \\ y_i = J & \text{if } y_i^* > \alpha_J \end{array} \right.$$

The α 's are unknown parameters to be estimated with β .

$$\alpha_1 < \alpha_2 < \dots < \alpha_J$$

As before, we assume that ε is normally distributed across observations. Other distributions, particularly the logistic, could be used just easily. We assume the normal purely for convenience. The logistic and normal distributions generally give similar results in practice.

Alternatives to proportional odds model

- 1 Generalized ordered logit model
- 2 The stereotype model
- 3 The continuation ratio model
- 4 Cumulative Probit Models
- 5 Cumulative Log-Log Links

goodness-of-fit tests for ordinal models

Fagerland MW, Hosmer DW. A goodness-of-fit test for the proportional odds regression model., *Statistics in Medicine* 2013;32:2235-2249.

- For ordinal regression models, few methods exist to assess goodness of fit, and no general goodness-of-fit test is widely available in software packages.
- Lipsitz et al. proposed a goodness-of-fit test, partly based on the Hosmer–Lemeshow approach, for a general ordinal regression model. The test provides no contingency table of observed and estimated frequencies and is not always computable for small sample sizes.
- Pulkstenis and Robinson modified the Pearson χ^2 and deviance statistics such that they can be used for testing goodness of fit in ordinal models that include continuous covariates. One limitation with that approach is that the regression model needs to include both continuous and categorical covariates. Also, if many categorical covariates are present, the performance of the tests may suffer because of the large number of cells from which to calculate the Pearson χ^2 and deviance statistics.
- For practical application, we thereby recommend calculation of all three methods

- ① The end of this quick intro
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Ordered Choice Models

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Why?

Example 1

Let's assume that we observe variable y_i that codes customers' satisfaction from a certain product.

$$y_i \in \{\text{dissatisfied, neutral, satisfied}\} \quad (6)$$

Let's assign values (0, 1, 2) to the customers' satisfaction levels.

$$y_i = \beta \text{age}_i + \varepsilon_i \quad \varepsilon_i \sim N(0, 1) \quad (7)$$

Determine marginal effects in this models.

Exercise 2 from Advanced Econometrics Exam (2015)

Lecture given in Polish. Copyright©2015 dr Jerzy Mycielski.

Let's consider what factors determine economic migration. In her research, an analyst used data from Social Diagnosis Surveys from years 2000, 2003, 2005 and 2007. In the table below, you may find a probit estimates. What is more, the marginal effects were calculated for average characteristics. The dependent variable was binary and it took 1 as a value when a person worked abroad while the survey was conducted and 0 otherwise. Logarithm of income *per capita*, years of education, binary variable for sex, and binary variables for years constituted the set of explanatory variables.

Exercise from an exam

	parameter	marginal effect	z	p
ln(income)	-0.292	-0.004	-8.387	0.000
years_educ	0.064	0.001	8.332	0.000
female	-0.334	-0.005	-7.417	0.000
2003	-0.118	-0.002	-1.240	0.215
2005	0.388	0.008	4.885	0.000
2007	0.598	0.012	7.958	0.000
Constant	-1.550	-7.544		0.000
N	38,916			
R^2 McKelvey-Zavoina	0.134			
Count- R^2	0.992			
Adj. Count- R^2	0.000			

Questions

- a) What are problems will we encounter estimating linear probability model for the problem?
- b) What can we conclude using estimates?
- c) Interpret R^2 statistics.
- d) Interpret marginal effect for sex (0 for males, 1 for females).
- e) Interpret marginal effect for income.
- f) What model should be estimated to test the hypothesis that education impacts decision for working abroad differently on males and females?
- g) The hypothesis that years do not impact the decision of economic emigration was tested. The LR test statistic was equal to $LR_{test} = 154.95$. Verify this hypothesis.

Hint: $\chi_1^2(0.95) = 3.84$, $\chi_2^2(0.95) = 5.99$, $\chi_3^2(0.95) = 7.81$.

Bibliography

- [1] J. Campos, N.R. Ericsson, D.F. Hendry, *General-to-specific Modeling: An Overview and Selected Bibliography*, Board of Governors of the Federal Reserve System, International Finance Discussion Papers, Number 838, August 2005.
- [2] J. Mycielski, *Ekonometria*, textbook for WNE UW students [in Polish], University of Warsaw, 2009.
- [3] Hendry D.F., *Econometric Methodology: A Personal Perspective*, Chpter 10 in T.F. Bewley (ed.) *Advances in Econometrics: Fifth World Congress*, Volume 2, Cambridge University Press, Cambridge, 29-48.